

Hypothesis tests – HPT (Czech data)

Multilevel models for H1-H2

HPT and Extremism project

2026-10-02

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1 1. Purpose and hypotheses

This document presents multilevel association models relevant to the registered hypotheses H1 and H2. It does not implement the exact registered one-tailed, covariate-adjusted specifications and therefore is not a confirmatory test of those hypotheses; Table S5 documents the deviations.

- **H1.** Higher right-authoritarian / pro-Nazi attitudes predict **higher HPT scores** on the original instrument (risk of ideological contamination). Predictors: FR-LF-mini (total or RD/NS facets) and KSA-3.
- **H2.** The H1 effect **persists controlling** for prior knowledge (KN total) and social desirability (SDR-5).

Notes on constructs and scoring:

- HPT subscores (POP, ROA, CONT) follow Hartmann & Hasselhorn / Huijgen et al. We treat **POP items as presentist** and therefore **reverse-score POP** so that **higher = more contextualised reasoning**. DVs used here are **HPT total (CTX6: POP_rev+CONT)**, **CONT**, and **POP_rev**.
- FR-LF-mini uses **RD1-RD3** and **NS1-NS3**; we analyse **total** and **RD/NS facets**.
- KSA-3 (9 items; aggression, submission, conventionalism) is included as a convergent authoritarian predictor. (Registered.)

2 2. Data, variables, and preprocessing

```
# Core
library(tidyverse)
library(readxl)
library(janitor)

# Models + tables
library(lme4)
library(lmerTest)
library(performance)
library(effectsize)
library(broom.mixed)
library(modelsummary)
library(glue)

library(kableExtra)
options(
  modelsummary_format = "latex",
  modelsummary_factory_latex = "kableExtra"
)
source("submissions/pci_psychology/scoring_helpers.R")

# Load the dataset created in 00_data-preparation
load("normalised_responses.RData")
stopifnot(exists("normalised_responses"))

# Clean names to lower_snake so items are pop1/roa1/cont1 etc.
dat_raw <- normalised_responses |> janitor::clean_names()

# -----
# Build a UNIQUE class identifier = school_id x class label
# We support multiple plausible column names from the codebook.
# -----
# Detect school id column
school_var <- names(dat_raw)[names(dat_raw) %in% c("school_id", "school")]
# Detect class label column (human-readable class label)
class_label_var <- names(dat_raw)[names(dat_raw) %in% c("classroom_label", "class_label", "class")]

if (length(school_var) == 0) stop("No school id column found (tried: school_id, school).")
if (length(class_label_var) == 0) stop("No class label column found (tried: classroom_label, class_label, class).")
```

```

school_var <- school_var[1]
class_label_var <- class_label_var[1]

# Force factors and create class_id
dat_raw <- dat_raw |>
  mutate(
    !!school_var := as.factor(.data[[school_var]]),
    !!class_label_var := as.factor(.data[[class_label_var]]),
    class_id = interaction(.data[[school_var]], .data[[class_label_var]], drop = TRUE)
  )

# -----
# HPT item vectors (lowercase after clean_names())
# -----
pop_items <- paste0("pop", 1:3)
roa_items <- paste0("roa", 1:3)
cont_items <- paste0("cont", 1:3)

# Reverse POP items so higher = more contextualised (1-4 scale assumed)
dat_raw <- dat_raw %>%
  mutate(across(all_of(pop_items), ~ 5 - as.numeric(.), .names = "{.col}_rev"))

# ---- Knowledge ----
kn_items <- paste0("kn", 1:6)

dat <- dat_raw %>%
  mutate(
    kn_total = rowSums(across(all_of(kn_items)), na.rm = TRUE)
  )

# ---- HPT (use reversed POP) ----
dat <- dat %>%
  mutate(
    hpt_pop_rev = scale_mean(., paste0(pop_items, "_rev"), min_answered = 2),
    hpt_cont     = scale_mean(., cont_items, min_answered = 2),
    hpt_roa      = scale_mean(., roa_items, min_answered = 2),
    # Primary total (CTX6 = POP_rev + CONT); keep 9-item as sensitivity if needed
    hpt_total    = rowMeans(cbind(hpt_pop_rev, hpt_cont), na.rm = FALSE),
    hpt_total9   = rowMeans(cbind(hpt_pop_rev, hpt_cont, hpt_roa),
                          na.rm = FALSE)
  )

```

```

# ---- FR-LF mini ----
rd_items <- paste0("rd", 1:3)
ns_items <- paste0("ns", 1:3)

dat <- dat %>%
  mutate(
    frlf_rd = scale_mean(., rd_items, min_answered = 2),
    frlf_ns = scale_mean(., ns_items, min_answered = 2),
    frlf_tot = scale_mean(., c(rd_items, ns_items), min_answered = 4)
  )

# ---- KSA-3 ----
a_items <- paste0("a", 1:3)
u_items <- paste0("u", 1:3)
k_items <- paste0("k", 1:3)
ksa_items <- c(a_items, u_items, k_items)

dat <- dat %>%
  mutate(
    ksa3_a = scale_mean(., a_items, min_answered = 2),
    ksa3_u = scale_mean(., u_items, min_answered = 2),
    ksa3_k = scale_mean(., k_items, min_answered = 2),
    ksa3_tot = scale_mean(., ksa_items, min_answered = 7)
  )

# ---- SDR-5 ----
sdr_items <- paste0("sdr", 1:5)

dat <- dat %>%
  mutate(
    sdr5_tot = scale_mean(., sdr_items, min_answered = 4)
  )

# Z-standardise continuous predictors (for comparability)
z <- function(x) as.numeric(scale(x))

# Ensure clustering vars present for every analysed row

dat <- dat |>
  mutate(
    z_hpt_total = z(hpt_total),
    z_hpt_cont = z(hpt_cont),

```

```

z_hpt_pop    = z(hpt_pop_rev),

z_frlf_tot = z(frlf_tot),
z_frlf_rd  = z(frlf_rd),
z_frlf_ns  = z(frlf_ns),

z_ksa3_tot = z(ksa3_tot),

z_kn_total = z(kn_total),
z_sdr5_tot = z(sdr5_tot)
) |>
drop_na(all_of(c(school_var, "class_id")))

```

3 3. Model plan

We estimate **random-intercept multilevel models** with **two clustering terms** (students nested in classes within schools):

- Base (FR-LF total): $DV \sim z_frlf_tot + z_ksa3_tot + z_kn_total + z_sdr5_tot + (1 | school_id) + (1 | class_id)$
- Facet (RD/NS): $DV \sim z_frlf_rd + z_frlf_ns + z_ksa3_tot + z_kn_total + z_sdr5_tot + (1 | school_id) + (1 | class_id)$
- Interaction (if preregistered): $DV \sim z_frlf_tot * z_kn_total + z_ksa3_tot + z_sdr5_tot + (1 | school_id) + (1 | class_id)$

DVs: z_hpt_total (CTX6), z_hpt_cont, z_hpt_pop (POP_rev).

```

dv_list <- c("z_hpt_total", "z_hpt_cont", "z_hpt_pop")

fits <- list()

for (dv in dv_list) {
  form_base <- as.formula(
    glue("{dv} ~ z_frlf_tot + z_ksa3_tot + z_kn_total + z_sdr5_tot + (1 | {school_var}) + (1 | class_id)")
  )
  form_facet <- as.formula(
    glue("{dv} ~ z_frlf_rd + z_frlf_ns + z_ksa3_tot + z_kn_total + z_sdr5_tot + (1 | {school_var}) + (1 | class_id)")
  )
  form_int <- as.formula(
    glue("{dv} ~ z_frlf_tot * z_kn_total + z_ksa3_tot + z_sdr5_tot + (1 | {school_var}) + (1 | class_id)")
  )

  m_base <- lmer(form_base, data = dat)
  m_facet <- lmer(form_facet, data = dat)
  m_int <- lmer(form_int, data = dat)

  fits[[dv]] <- list(base=m_base, facet=m_facet, int=m_int)
}

```

	HPT total (CTX6) – Base	HPT total (CTX6) – Facet	HPT total (CTX6) – Int.
(Intercept)	–0.019 (0.068)	–0.021 (0.069)	–0.020 (0.067)
z_frlf_tot	0.016 (0.068)		0.017 (0.068)
z_ksa3_tot	–0.061 (0.067)	–0.056 (0.067)	–0.061 (0.067)
z_kn_total	0.340*** (0.058)	0.341*** (0.058)	0.337*** (0.058)
z_sdr5_tot	–0.040 (0.058)	–0.029 (0.059)	–0.039 (0.058)
z_frlf_rd		–0.045 (0.067)	
z_frlf_ns		0.065 (0.066)	
z_frlf_tot × z_kn_total			–0.019 (0.060)
SD (Intercept class_id)	0.000	0.000	0.000
SD (Intercept school_id)	0.096	0.100	0.092
SD (Observations)	0.944	0.944	0.946
Num.Obs.	282	282	282
R2 Marg.	0.115	0.118	0.115
RMSE	0.93	0.93	0.93

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```
}
```

```
msummary(
  list(
    "HPT total (CTX6) -- Base" = fits$z_hpt_total$base,
    "HPT total (CTX6) -- Facet" = fits$z_hpt_total$facet,
    "HPT total (CTX6) -- Int." = fits$z_hpt_total$int
  ),
  statistic = "{std.error}",
  gof_omit = "IC|Log|AIC|BIC",
  stars = TRUE
)
```

```
msummary(
  list(
```

	CONT – Base	CONT – Facet	CONT – Int.
(Intercept)	−0.023 (0.073)	−0.024 (0.073)	−0.030 (0.071)
z_frlf_tot	0.006 (0.069)		0.011 (0.069)
z_ksa3_tot	0.054 (0.069)	0.061 (0.069)	0.057 (0.069)
z_kn_total	0.215*** (0.059)	0.218*** (0.059)	0.201*** (0.060)
z_sdr5_tot	0.011 (0.059)	0.025 (0.060)	0.015 (0.059)
z_frlf_rd		−0.075 (0.068)	
z_frlf_ns		0.082 (0.067)	
z_frlf_tot × z_kn_total			−0.097 (0.061)
SD (Intercept class_id)	0.114	0.123	0.121
SD (Intercept school_id)	0.088	0.086	0.075
SD (Observations)	0.966	0.964	0.964
Num.Obs.	282	282	282
R2 Marg.	0.050	0.057	0.059
R2 Cond.	0.071	0.079	0.079
RMSE	0.95	0.95	0.95

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```

"CONT -- Base" = fits$z_hpt_cont$base,
"CONT -- Facet" = fits$z_hpt_cont$facet,
"CONT -- Int." = fits$z_hpt_cont$int
),
statistic = "({std.error})",
gof_omit = "IC|Log|AIC|BIC",
stars = TRUE
)

```

```

msummary(
  list(
    "POP_rev -- Base" = fits$z_hpt_pop$base,
    "POP_rev -- Facet" = fits$z_hpt_pop$facet,
    "POP_rev -- Int." = fits$z_hpt_pop$int
  )
)

```

	POP_rev – Base	POP_rev – Facet	POP_rev – Int.
(Intercept)	–0.007 (0.056)	–0.007 (0.056)	0.001 (0.056)
z_frlf_tot	0.011 (0.067)		0.006 (0.067)
z_ksa3_tot	–0.166* (0.066)	–0.166* (0.067)	–0.168* (0.066)
z_kn_total	0.335*** (0.057)	0.335*** (0.057)	0.346*** (0.057)
z_sdr5_tot	–0.080 (0.057)	–0.079 (0.058)	–0.084 (0.057)
z_frlf_rd		0.002 (0.066)	
z_frlf_ns		0.012 (0.066)	
z_frlf_tot × z_kn_total			0.075 (0.059)
SD (Intercept class_id)	0.000	0.000	0.000
SD (Intercept school_id)	0.000	0.000	0.000
SD (Observations)	0.940	0.942	0.939
Num.Obs.	282	282	282
R2 Marg.	0.131	0.131	0.135
RMSE	0.93	0.93	0.93

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

```
),
  statistic = "{std.error}",
  gof_omit = "IC|Log|AIC|BIC",
  stars = TRUE
)
```

```
`%||%' <- function(a, b) if (!is.null(a) && length(a) > 0) a else b
```

```
collect_metrics <- function(m) {
  icc_val <- tryCatch({
    ic <- performance::icc(m)
    as.numeric(ic$ICC_adjusted %||% ic$ICC %||% ic$ICC_conditional %||% NA_real_)
  }, error = function(e) NA_real_)

```

```
  r2m <- r2c <- NA_real_
}
```

```

try({
  r2o <- performance::r2_nakagawa(m)
  r2m <- as.numeric(r2o$R2_marginal %||% r2o$R2m %||% NA_real_)
  r2c <- as.numeric(r2o$R2_conditional %||% r2o$R2c %||% NA_real_)
}, silent = TRUE)

data.frame(ICC = icc_val, R2_m = r2m, R2_c = r2c, check.names = FALSE)
}

metrics <- dplyr::bind_rows(
  list(
    `HPT total (CTX6) -- Base` = collect_metrics(fits$z_hpt_total$base),
    `HPT total (CTX6) -- Facet` = collect_metrics(fits$z_hpt_total$facet),
    `HPT total (CTX6) -- Int.` = collect_metrics(fits$z_hpt_total$int),
    `CONT -- Base` = collect_metrics(fits$z_hpt_cont$base),
    `CONT -- Facet` = collect_metrics(fits$z_hpt_cont$facet),
    `CONT -- Int.` = collect_metrics(fits$z_hpt_cont$int),
    `POP_rev -- Base` = collect_metrics(fits$z_hpt_pop$base),
    `POP_rev -- Facet` = collect_metrics(fits$z_hpt_pop$facet),
    `POP_rev -- Int.` = collect_metrics(fits$z_hpt_pop$int)
  ),
  .id = "Model"
)

```

```

## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
## Random effect variances not available. Returned R2 does not account for random effects.
knitr::kable(metrics, digits = 3, caption = "Model fit and clustering (ICC,  $R^2$ ).")

```

Table 1: Model fit and clustering (ICC, R^2).

Model	ICC	R2_m	R2_c
HPT total (CTX6) – Base	NA	0.115	NA
HPT total (CTX6) – Facet	NA	0.118	NA
HPT total (CTX6) – Int.	NA	0.115	NA
CONT – Base	0.022	0.050	0.071

Model	ICC	R2_m	R2_c
CONT – Facet	0.024	0.057	0.079
CONT – Int.	0.021	0.059	0.079
POP_rev – Base	NA	0.131	NA
POP_rev – Facet	NA	0.131	NA
POP_rev – Int.	NA	0.135	NA

```

tidy_all <- function(lst, label) {
  bind_rows(
    broom.mixed::tidy(lst$base, effects="fixed", conf.int=TRUE) |> mutate(spec="Base"),
    broom.mixed::tidy(lst$facet, effects="fixed", conf.int=TRUE) |> mutate(spec="Facet"),
    broom.mixed::tidy(lst$int, effects="fixed", conf.int=TRUE) |> mutate(spec="Interaction")
  ) |>
  filter(term != "(Intercept)") |>
  mutate(dv = label)
}

tidy_tbl <- bind_rows(
  tidy_all(fits$z_hpt_total, "HPT total (CTX6)"),
  tidy_all(fits$z_hpt_cont, "CONT"),
  tidy_all(fits$z_hpt_pop, "POP_rev")
)

knitr::kable(
  tidy_tbl |> select(dv, spec, term, estimate, conf.low, conf.high, p.value),
  digits = 3,
  caption = "Fixed effects (standardized coefficients)."
```

Table 2: Fixed effects (standardized coefficients).

dv	spec	term	estimate	conf.low	conf.high	p.value
HPT total (CTX6)	Base	z_frlf_tot	0.016	-0.117	0.149	0.817
HPT total (CTX6)	Base	z_ksa3_tot	-0.061	-0.193	0.071	0.362
HPT total (CTX6)	Base	z_kn_total	0.340	0.226	0.453	0.000
HPT total (CTX6)	Base	z_sdr5_tot	-0.040	-0.153	0.074	0.492
HPT total (CTX6)	Facet	z_frlf_rd	-0.045	-0.176	0.087	0.503
HPT total (CTX6)	Facet	z_frlf_ns	0.065	-0.065	0.194	0.327
HPT total (CTX6)	Facet	z_ksa3_tot	-0.056	-0.189	0.076	0.402
HPT total (CTX6)	Facet	z_kn_total	0.341	0.228	0.455	0.000

dv	spec	term	estimate	conf.low	conf.high	p.value
HPT total (CTX6)	Facet	z_sdr5_tot	-0.029	-0.145	0.086	0.618
HPT total (CTX6)	Interaction	z_frlf_tot	0.017	-0.117	0.150	0.806
HPT total (CTX6)	Interaction	z_kn_total	0.337	0.222	0.452	0.000
HPT total (CTX6)	Interaction	z_ksa3_tot	-0.061	-0.193	0.071	0.367
HPT total (CTX6)	Interaction	z_sdr5_tot	-0.039	-0.153	0.075	0.503
HPT total (CTX6)	Interaction	z_frlf_tot:z_kn_total	-0.019	-0.136	0.099	0.755
CONT	Base	z_frlf_tot	0.006	-0.130	0.143	0.929
CONT	Base	z_ksa3_tot	0.054	-0.082	0.189	0.436
CONT	Base	z_kn_total	0.215	0.099	0.332	0.000
CONT	Base	z_sdr5_tot	0.011	-0.106	0.127	0.859
CONT	Facet	z_frlf_rd	-0.075	-0.209	0.060	0.275
CONT	Facet	z_frlf_ns	0.082	-0.051	0.215	0.225
CONT	Facet	z_ksa3_tot	0.061	-0.075	0.197	0.375
CONT	Facet	z_kn_total	0.218	0.102	0.334	0.000
CONT	Facet	z_sdr5_tot	0.025	-0.093	0.144	0.671
CONT	Interaction	z_frlf_tot	0.011	-0.125	0.148	0.868
CONT	Interaction	z_kn_total	0.201	0.084	0.319	0.001
CONT	Interaction	z_ksa3_tot	0.057	-0.079	0.192	0.410
CONT	Interaction	z_sdr5_tot	0.015	-0.101	0.132	0.799
CONT	Interaction	z_frlf_tot:z_kn_total	-0.097	-0.217	0.023	0.114
POP_rev	Base	z_frlf_tot	0.011	-0.120	0.143	0.867
POP_rev	Base	z_ksa3_tot	-0.166	-0.297	-0.036	0.013
POP_rev	Base	z_kn_total	0.335	0.223	0.447	0.000
POP_rev	Base	z_sdr5_tot	-0.080	-0.193	0.032	0.162
POP_rev	Facet	z_frlf_rd	0.002	-0.128	0.133	0.970
POP_rev	Facet	z_frlf_ns	0.012	-0.117	0.141	0.856
POP_rev	Facet	z_ksa3_tot	-0.166	-0.297	-0.035	0.013
POP_rev	Facet	z_kn_total	0.335	0.223	0.447	0.000
POP_rev	Facet	z_sdr5_tot	-0.079	-0.194	0.035	0.174
POP_rev	Interaction	z_frlf_tot	0.006	-0.125	0.138	0.924
POP_rev	Interaction	z_kn_total	0.346	0.233	0.459	0.000
POP_rev	Interaction	z_ksa3_tot	-0.168	-0.298	-0.038	0.011
POP_rev	Interaction	z_sdr5_tot	-0.084	-0.196	0.029	0.145
POP_rev	Interaction	z_frlf_tot:z_kn_total	0.075	-0.042	0.191	0.209

4 4. Results – decision rules

Interpret these models as associations related to the registered hypotheses, not as the exact confirmatory tests; see Table S5 for the deviations:

- An **H1-relevant association** has a positive coefficient for **FR-LF** (either `z_frlf_tot` in Base/Int. or `z_frlf_rd/z_frlf_ns` in Facet) with $p < .05$ for

HPT total (CTX6) and/or **CONT**.

- An **H2-relevant controlled association** has the same pattern after adding controls (**KN**, **SDR-5**) and **KSA-3** (already included). These are not the exact registered decision rules; see Table S5.

Reading POP_rev. Because POP is reversed, higher **POP_rev** means **less presentism / more contextualised fit** on items that originally cued presentist endorsements. Interpret alongside **CONT**.

5 5. Brief interpretation guide (for the write-up)

- **Effect size:** Coefficients are **standardised** (). Values around 0.10 are small, 0.20-0.30 moderate for individual-level predictors in multilevel models; report 95% CIs.
- **Clustering:** Report **ICC** to show class-level variance.
- **Model fit:** Report marginal and conditional R^2 and compare Base vs. Facet vs. Interaction.
- **Substantive meaning:** A **positive FR-LF** effect on **HPT total / CONT** suggests that ideological affinity **elevates apparent contextualisation**, consistent with the contamination concern.
- **Controls:** If FR-LF remains significant after **KN** and **SDR-5**, state that results are **not explained** by prior knowledge or social desirability (per H2).

6 6. Transparency and provenance

- HPT structure and reversal logic follow Hartmann & Hasselhorn / Huijgen et al.
- FR-LF-mini originates from the Leipzig FR-LF.
- Analysis plan: random-intercept LMMs; DVs: HPT total (CTX6), CONT, POP_rev; predictors: FR-LF (total; RD/NS facets), KSA-3; controls: KN, SDR-5; clustering: school + class_id.

7 7. Session info

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 24.04.5 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0 LAPACK version 3.12.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
##  [3] LC_TIME=cs_CZ.UTF-8      LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=cs_CZ.UTF-8  LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=cs_CZ.UTF-8     LC_NAME=C
```

```

## [9] LC_ADDRESS=C                LC_TELEPHONE=C
## [11] LC_MEASUREMENT=cs_CZ.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Prague
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] kableExtra_1.4.1    glue_1.8.1          modelsummary_2.6.0
## [4] broom.mixed_0.2.9.7 effectsize_1.0.3    performance_0.18.2
## [7] lmerTest_3.2-1      lme4_2.0-6          Matrix_1.7-6
## [10] janitor_2.2.1       readxl_1.5.0.1      lubridate_1.9.5
## [13] forcats_1.0.1       stringr_1.6.0       dplyr_1.2.1
## [16] purrr_1.2.2         readr_2.2.0         tidyr_1.3.2
## [19] tibble_3.3.1        ggplot2_4.0.3       tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1    viridisLite_0.4.3   farver_2.1.2
## [4] S7_0.2.2            fastmap_1.2.0       bayestestR_0.19.0
## [7] digest_0.6.39       timechange_0.4.0    lifecycle_1.0.5
## [10] magrittr_2.0.5      compiler_4.6.1      rlang_1.3.0
## [13] tools_4.6.1         yaml_2.3.12         data.table_1.18.6.1
## [16] knitr_1.51          xml2_1.6.0          RColorBrewer_1.1-3
## [19] tinytable_0.19.0    withr_3.0.3         numDeriv_2016.8-1.1
## [22] grid_4.6.1          datawizard_1.4.0    future_1.75.0
## [25] globals_0.19.1     scales_1.4.0        MASS_7.3-66
## [28] insight_1.5.4       cli_3.6.6           rmarkdown_2.32
## [31] reformulas_0.4.4    generics_0.1.4      otel_0.2.0
## [34] future.apply_1.20.2 rstudioapi_0.19.0   tzdb_0.5.0
## [37] parameters_0.29.3   minqa_1.2.8         splines_4.6.1
## [40] parallel_4.6.1      cellranger_1.1.0    vctrs_0.7.3
## [43] boot_1.3-32         hms_1.1.4           listenv_1.0.0
## [46] systemfonts_1.3.2   parallelly_1.48.0   nloptr_2.2.1
## [49] codetools_0.2-20    stringi_1.8.7       gtable_0.3.6
## [52] tables_0.9.35       pillar_1.11.1       furrr_0.4.0
## [55] htmltools_0.5.9     R6_2.6.1            textshaping_1.0.5
## [58] Rdpack_2.6.6        evaluate_1.0.5      lattice_0.23-1
## [61] rbibutils_2.4.1     backports_1.5.1     broom_1.0.13
## [64] snakecase_0.11.1    Rcpp_1.1.2          checkmate_2.3.4
## [67] svglite_2.2.2       nlme_3.1-171        xfun_0.60

```

```
## [70] pkgconfig_2.0.3
```